

Machine Learning in Python

Neural Networks I: Foundations and Optimization

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Outline

- 1 Introduction to Artificial Neural Networks
- 2 The Multilayer Perceptron (MLP)
- 3 Training Neural Networks
- 4 Optimizing Neural Networks
- 5 Building Networks in PyTorch
- 6 Summary

What is an Artificial Neural Network?

Definition

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Key Properties

Example

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- Foundation of **Deep Learning**
- Can learn complex, non-linear relationships
- Powers image recognition, voice assistants, medical diagnosis, and more

Key Components of a Neural Network

The Building Blocks

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- ❹ **Activation Function**: Introduces non-linearity, allowing the network to learn curves

Biological Inspiration

From Biology to Mathematics

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ANNs are loosely inspired by biological neural networks in the brain:

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ANNs are loosely inspired by biological neural networks in the brain:

Biological Neuron	Artificial Neuron	Role
Dendrites	Inputs (x)	Receives signals
Synapse	Weights (w)	Controls signal strength
Cell Body	Summation ($\sum$)	Aggregates signals
Axon	Activation	Transmits output

The Perceptron (1958)

The Simplest Neural Network

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Mathematical Formulation

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Mathematical Formulation

① **Weighted Sum:** $z = w_1x_1 + w_2x_2 + \dots + b$

② **Activation:**

$$\text{Output} = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z \leq 0 \end{cases}$$

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This can solve **linearly separable** problems (drawing a straight line between classes).

The XOR Problem

The Fatal Flaw of Single Perceptrons

The Limitation

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The XOR Problem

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x_1	x_2	AND	XOR
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0	1	0	1
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- **AND Gate**: Linearly separable (one line can separate classes)
- **XOR Gate**: NOT linearly separable (needs a curve or multiple lines)
- This limitation highlighted the need for **Multi-Layer Perceptrons (MLPs)** and better training methods

From One to Many

Stacking Neurons

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Architecture

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A **Multilayer Perceptron (MLP)** or **Feedforward Neural Network** connects many neurons in layers to solve complex, non-linear problems.

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- ➌ **Output Layer:** Produces final prediction (e.g., probability of class)

Activation Functions

Why Activation Functions?

Common Activation Functions

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Without activation functions, an MLP is just a giant Linear Regression (linear + linear = linear). **Activation functions** introduce **non-linearity**, allowing networks to learn curves.

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Function	Formula	Range	Usage
Sigmoid	$\frac{1}{1+e^{-z}}$	$(0, 1)$	Output (Binary Class.)
Tanh	$\frac{e^z - e^{-z}}{e^z + e^{-z}}$	$(-1, 1)$	Hidden layers (Legacy)
ReLU	$\max(0, z)$	$[0, \infty)$	Standard for Hidden
Softmax	$\frac{e^{z_i}}{\sum e^{z_j}}$	$(0, 1)$	Output (Multiclass)

Forward Propagation

How Data Flows Through the Network

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This process repeats until the output layer. The magic: `np.dot` handles all connections automatically.

Network Capacity: Width vs Depth

Architectural Choices

Trade-offs

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Network Capacity: Width vs Depth

Architectural Choices

Architecture	Structure	Characteristics
Wide	1 layer, 100 neurons	Simple patterns, fast
Deep	5 layers, 10 neurons	Hierarchical features, complex

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- Deeper networks learn hierarchical representations (edges $\rightarrow$ parts $\rightarrow$ objects)
- Wider networks can approximate functions but may need more parameters
- In practice: start simple, add complexity if needed

The Goal of Training

From Random to Learned

Three Components

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A neural network starts with **random weights**. It knows nothing. **Training** adjusts these weights to minimize prediction error.

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- 2 **Optimizer**: How we update weights

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- 1 **Loss Function**: How we measure error
- 2 **Optimizer**: How we update weights
- 3 **Backpropagation**: How we calculate the updates

Loss Functions

The Scorecard

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- MSE: Penalizes large errors heavily (prices, temperatures)
- Cross-Entropy: Penalizes confident wrong answers (probabilities)

Stochastic Gradient Descent (SGD)

Optimization Strategy

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SGD updates weights after seeing just **one example** (or a small "mini-batch").

- **Pros:** Faster, noise helps escape local minima

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- **Cons:** Zigzaggy path to convergence

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The Chain Rule in Action

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How do we know which weight in Layer 1 caused the error in Layer 5? We use the **Chain Rule** of calculus.

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 - "Layer 3, based on Layer 4's error, you contributed Y..."

The Vanishing Gradient Problem

Why Deep Networks Failed (Initially)

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The Vanishing Gradient Problem

Why Deep Networks Failed (Initially)

In deep networks, gradients are multiplied repeatedly during backpropagation. **Sigmoid** has max derivative of 0.25:

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- In practice, combine with good initialization, normalization, and residual connections

The Learning Rate

The Most Critical Hyperparameter

The Goldilocks Zone

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Learning Rate	Behavior
Too Small	Loss decreases very slowly (0.0001)
Too Large	Loss oscillates or explodes (10.0)
Just Right	Stable descent (optimizer-dependent; e.g. Adam 1e-3, SGD 1e-2 to 1e-1)

Weight Initialization

Starting Right

Strategies

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- Both keep variance of activations constant across layers

Advanced Optimizers

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Adam is the default choice for most tasks.

Regularization: Dropout

Fighting Overfitting

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- At test time, all neurons are used (with scaled weights)

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- Enables training of much deeper networks

Early Stopping

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- Requires light tuning (mainly `patience`, optionally `min_delta`)

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Beyond Scikit-learn

PyTorch Advantages

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- Readable, "Pythonic" style
- Research and education standard
- Dynamic computation graphs (easy debugging)

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- `x.to("cuda")` — Move to GPU

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- `torch.ones(2, 3)` — Shape (2, 3) of ones
- `x.to("cuda")` — Move to GPU
- Math: `x * 2 + 5` — Same as NumPy

Defining a Network (`nn.Module`)

The Blueprint

Two Required Methods

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The 5-Step Boilerplate

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Wrap in a `for epoch in range(epochs):` loop.

PyTorch Outputs and Loss Pairings

Match Model Output to Loss

Inference Rule

Example

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- **Class label:** use threshold/argmax directly

Loss Functions and Optimizers

Loss Functions

Example

Task	Loss Class
Binary Classification	<code>nn.BCEWithLogitsLoss()</code>
Multiclass	<code>nn.CrossEntropyLoss()</code>
Regression	<code>nn.MSELoss()</code>

Optimizers

```
optimizer = torch.optim.Adam(model.parameters(), lr=0.01)
```

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```
model.eval()
with torch.no_grad():
    test_logits = model(X_test)
    predicted = (test_logits > 0).float()
    accuracy = (predicted == y_test).float().mean()
```

If you continue training afterward, switch back to `model.train()`.

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- 7 **PyTorch**: Tensors + nn.Module + Training Loop

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Neural Networks II: Deep Learning & Transformers

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